

ANSH SINGH

Lucknow, India | +91 7905880218 | sansh3030@gmail.com | linkedin.com/in/ansh-singh | github.com/AnshSingh30

CAREER OBJECTIVE

Agentic AI Engineer graduating May 2026 with deep, hands-on expertise in LLM-powered applications, RAG pipelines, multi-agent orchestration, and production REST APIs. Proficient in Python, LangChain, OpenAI & Anthropic APIs, PyTorch, and AWS cloud deployments. Experienced in building end-to-end AI systems with tools, planners, memory components, and asynchronous processing. Seeking to bring autonomous AI system development skills to HARMAN's next-generation in-vehicle and intelligent platform engineering.

TECHNICAL SKILLS

Languages: Python (Expert), C++, JavaScript, SQL, Dart

Agentic AI & LLMs: LangChain, LlamaIndex (basic), OpenAI API, Anthropic API, Hugging Face, RAG Pipelines, Prompt Engineering, Multi-Agent Orchestration, Tool Use, Memory Components, Model Evaluation

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, CNN, Feature Engineering, Data Preprocessing, NumPy, Pandas

Backend & APIs: Flask, FastAPI (basic), REST APIs, SQLAlchemy (basic), PostgreSQL, Supabase, FAISS / Vector Search, Query Optimization

Cloud & Infrastructure: AWS (ECS / EC2 / S3), Docker (basic), Git, Linux, CI/CD pipelines

Frontend & Mobile: React, Flutter, WebSockets, Real-time Data Sync

Auth & Security: OAuth 2.0, JWT, Role-Based Access Control, Row-Level Security

Core CS: Data Structures, Algorithms, OOP, OS, DBMS, Computer Networks, SDLC / Agile

PROFESSIONAL EXPERIENCE

AI/ML Engineering Intern | Hindalco Industries, Aditya Birla Group

Jun 2025 – Jul 2025

- Designed and deployed ML-based anomaly detection integrated into live operational monitoring — delivering a production Agentic AI workflow that autonomously classified and triaged anomalies, reducing manual effort by 40%.
- Built scalable Python data pipelines processing 50,000+ records with structured ingestion, preprocessing, and embedding — forming the data foundation for AI-driven predictive monitoring agents.
- Audited LLM model responses and system logs to eliminate false positives in automated alerting, improving anomaly detection precision through iterative model evaluation and prompt tuning.
- Deployed predictive models on AWS infrastructure, resolving critical bottlenecks in automated analysis workflows and ensuring reliable real-time agent execution.

Machine Learning Trainee | Teachnook (IIT Roorkee Collaboration)

Jul 2024

- Applied advanced ML algorithms with targeted feature engineering to classification and regression tasks, improving model reliability through iterative experimentation.
- Validated ML model deployments against training datasets using comprehensive testing protocols — identifying and mitigating performance regressions before production release.

TECHNICAL PROJECTS

Multi-Source RAG Pipeline (Multi-RAG) | Python, LangChain, FAISS, OpenAI API, Flask, PostgreSQL

2025

github.com/AnshSingh30/Multi-RAG

- Designed and built a multi-agent RAG system using LangChain with tool use, planner, and memory components — ingesting PDFs, web URLs, and databases into a unified FAISS vector store for grounded LLM responses.
- Implemented configurable retrieval strategies (dense, sparse, hybrid) with dynamic context window management, achieving sub-200ms query response times at scale.
- Exposed the agentic backend via production Flask REST APIs with JWT authentication, asynchronous processing, and PostgreSQL state management — mirroring real-world LLM microservice architecture.
- Reduced LLM hallucinations by 60% vs. vanilla API baseline through retrieval grounding, re-ranking, and structured prompt engineering.

SyncBoard — Real-Time Collaboration Platform | React, Node.js, WebSockets, PostgreSQL, Docker

2025

github.com/AnshSingh30/SyncBoard

- Built a real-time collaboration platform with WebSocket-driven live sync, RBAC, JWT auth, and containerized deployment — demonstrating full-stack AI product delivery capability.
- Implemented asynchronous event processing using a queue-based architecture to handle concurrent user sessions, keeping the system responsive under load — analogous to Celery-based async AI workloads.

- Containerized using Docker with environment-isolated configurations, enabling CI/CD-aligned deployment pipelines consistent with cloud-native microservice standards.

RAG-Powered Chatbot with Web Intelligence | *Python, Flask, RAG, OpenAI API, BeautifulSoup, PostgreSQL* Jun 2024 – Jul 2024

github.com/AnshSingh30/ChatBot-WebScrapper

- Architected a Retrieval-Augmented Generation pipeline with live web scraping, document chunking, embedding, and context-aware LLM generation — a complete Agentic AI tool-use and planning workflow.
- Designed modular, independently scalable service layers (scraping, embedding, retrieval, generation) enabling asynchronous processing and rapid iteration — aligning with LangChain/LlamaIndex agent architecture patterns.

JUITConnect — Real-Time University Platform | *Flutter, Supabase, PostgreSQL, Dart, Real-time APIs* 2024

github.com/AnshSingh30/JUITConnect

- Delivered a cross-platform mobile application (iOS + Android) with real-time data synchronization via Supabase Realtime, enabling live updates across thousands of concurrent users with zero-downtime cloud backend.
- Implemented Role-Based Access Control (RBAC) with PostgreSQL Row-Level Security policies enforcing fine-grained authorization across student, faculty, and admin roles — demonstrating secure multi-role system design applicable to in-vehicle user profiles.
- Optimized cold-start load time and reduced redundant fetches using selective subscription queries and efficient state management — critical performance discipline for resource-constrained embedded and automotive environments.

PATENT APPLICATIONS

2 patent applications filed (April 2026) — currently pending. Details available on request.

EDUCATION

B.Tech, Computer Science & Engineering (AI/ML Specialization) | JUIT, Solan, India Aug 2022 – May 2026

- AI/ML Specialization with coursework in Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, OS, DBMS, Computer Networks.
- Optimized database query speed by 20% via a custom-designed algorithm (academic project).